

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

School of Computer Science and Engineering

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 3 – Ultrasonic Sensor

Assoc Prof Nicholas Vun

More Specific Content Topics for Part 1

Topic 1: Electronic components & circuit interfacing
(related to Tutorial 1, Lab 1)

Topic 2 - SPI and I^2C Serial Interfaces
(related to Lab 1, Lab2)

Topic 3 – Ultrasonic Sensor
(related to Lab 1)

Topic 4 – IMU Sensor: Accelerometer & Gyroscope

Topic 5 – Sensor Fusion: Complementary Filter

Topic 6 – Kalman Filter

Topic 7 – Motors & Encoder Operations

Ultrasonics

Ultrasound

- sound whose frequency is too high to be heard by the human ear
- above 20 kHz frequency

Ultrasonic devices

- commonly used to detect objects and measure distances

Animal like Bats use a variety of ultrasonic ranging techniques to detect their preys

- using frequencies beyond 100 kHz

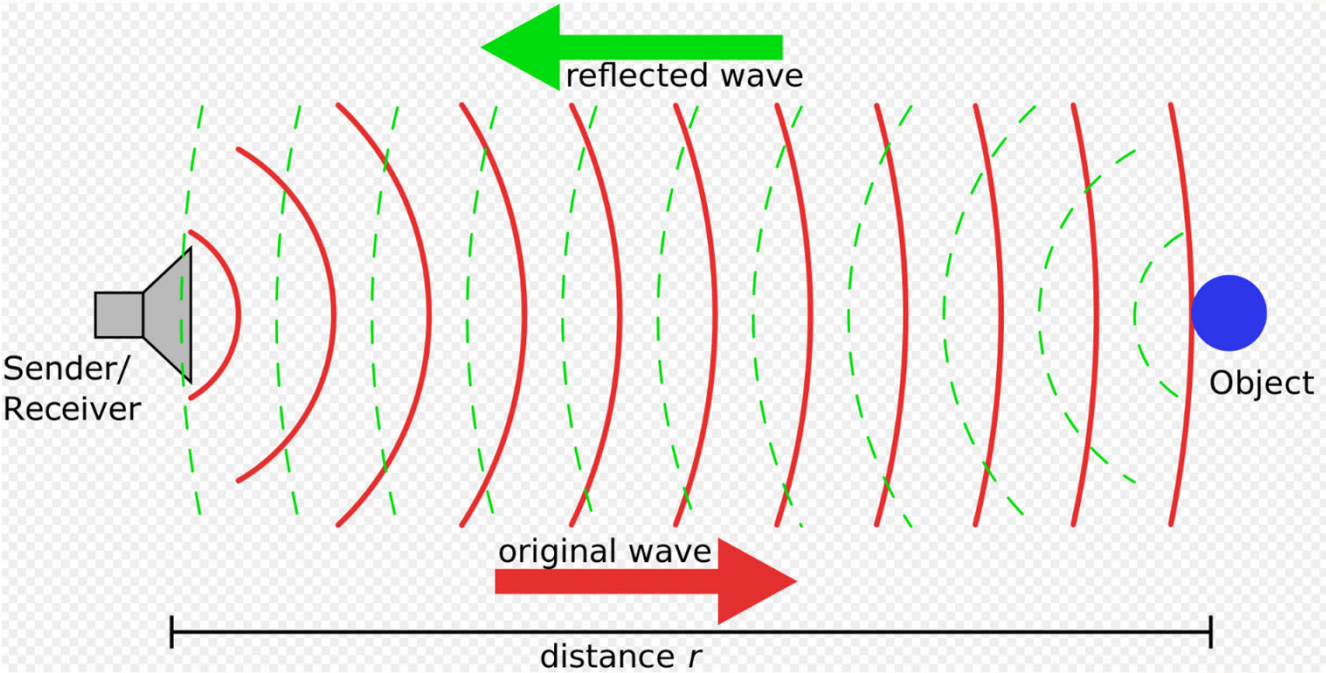
Ultrasonics Sensing

A common use of ultrasound

- finding distance

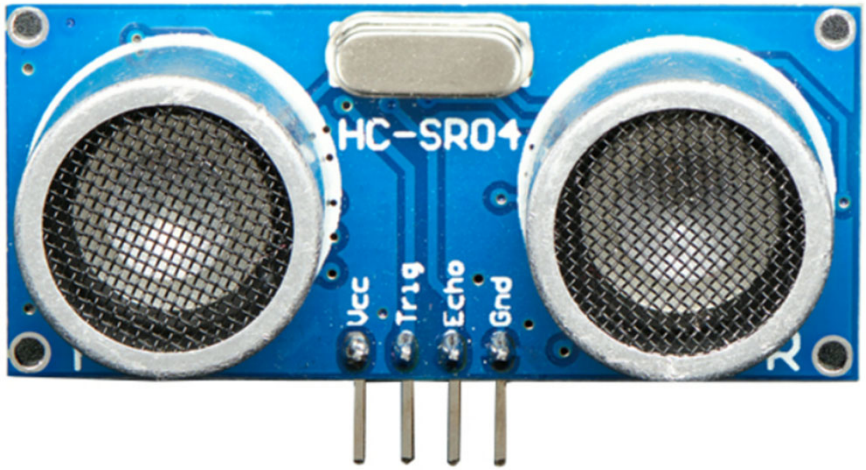
E.g., Underwater range finding – Sonar
Medical imaging - Sonogram

Principles of active sonic



(Source: Wikipedia)

Case Study: HC-SR04 Ultrasonic module



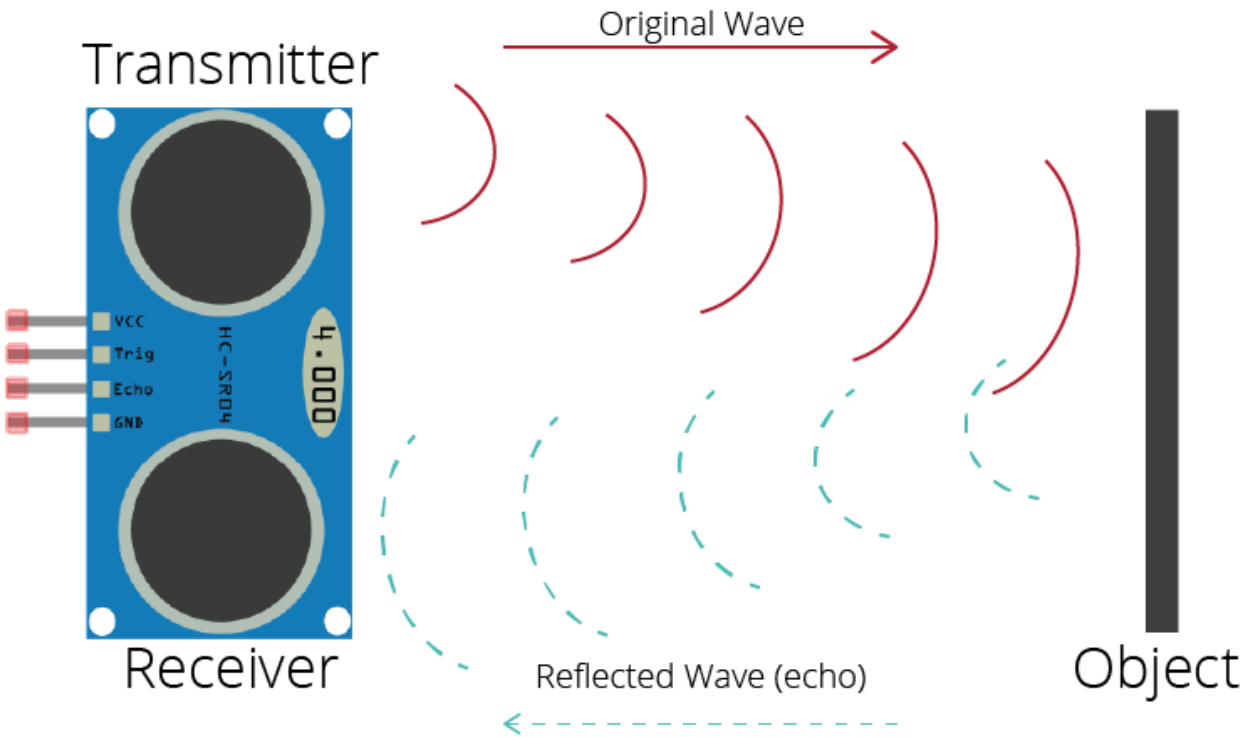
An ultrasonic sensing module

- measure the distance of an object located in front of the sensor
- within the range of 2cm to 4m

Operating Principles

Consists of two ultrasonic transducers

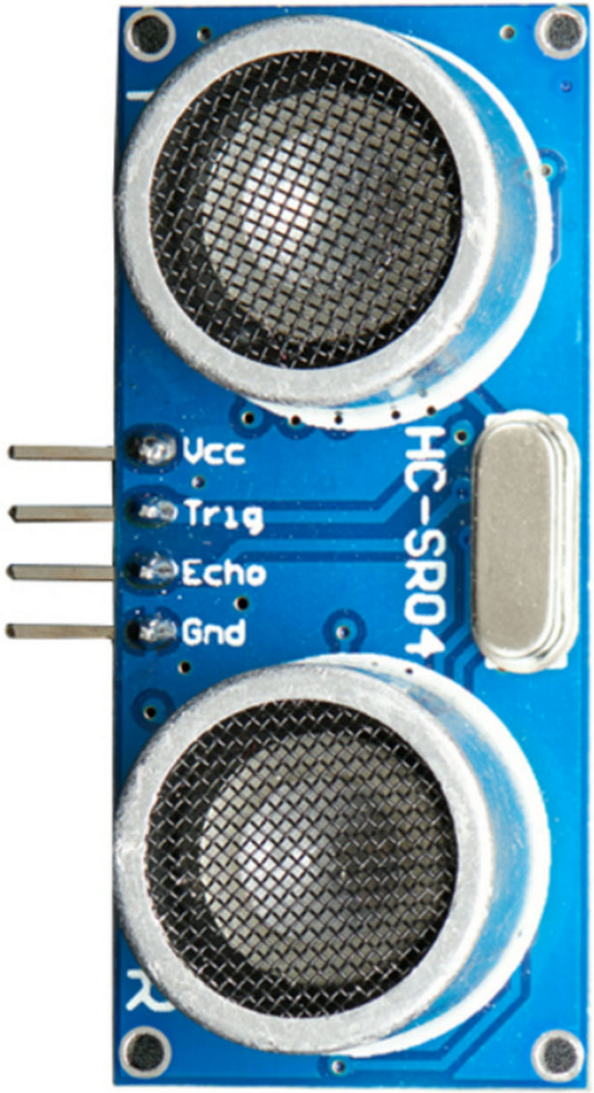
- one acts as a transmitter that converts trigger signal into 40 KHz ultrasonic sound pulses.
- the other acts as a receiver that listens for the transmitted pulses being reflected back

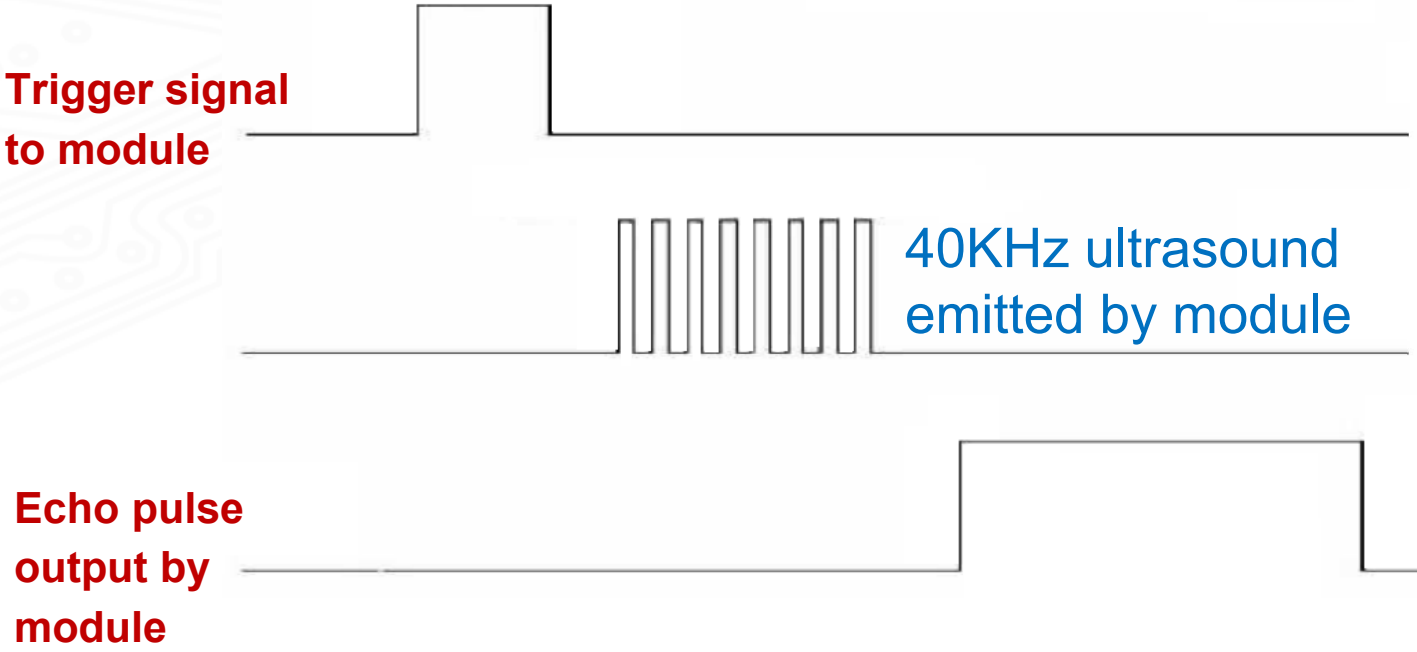


Operation

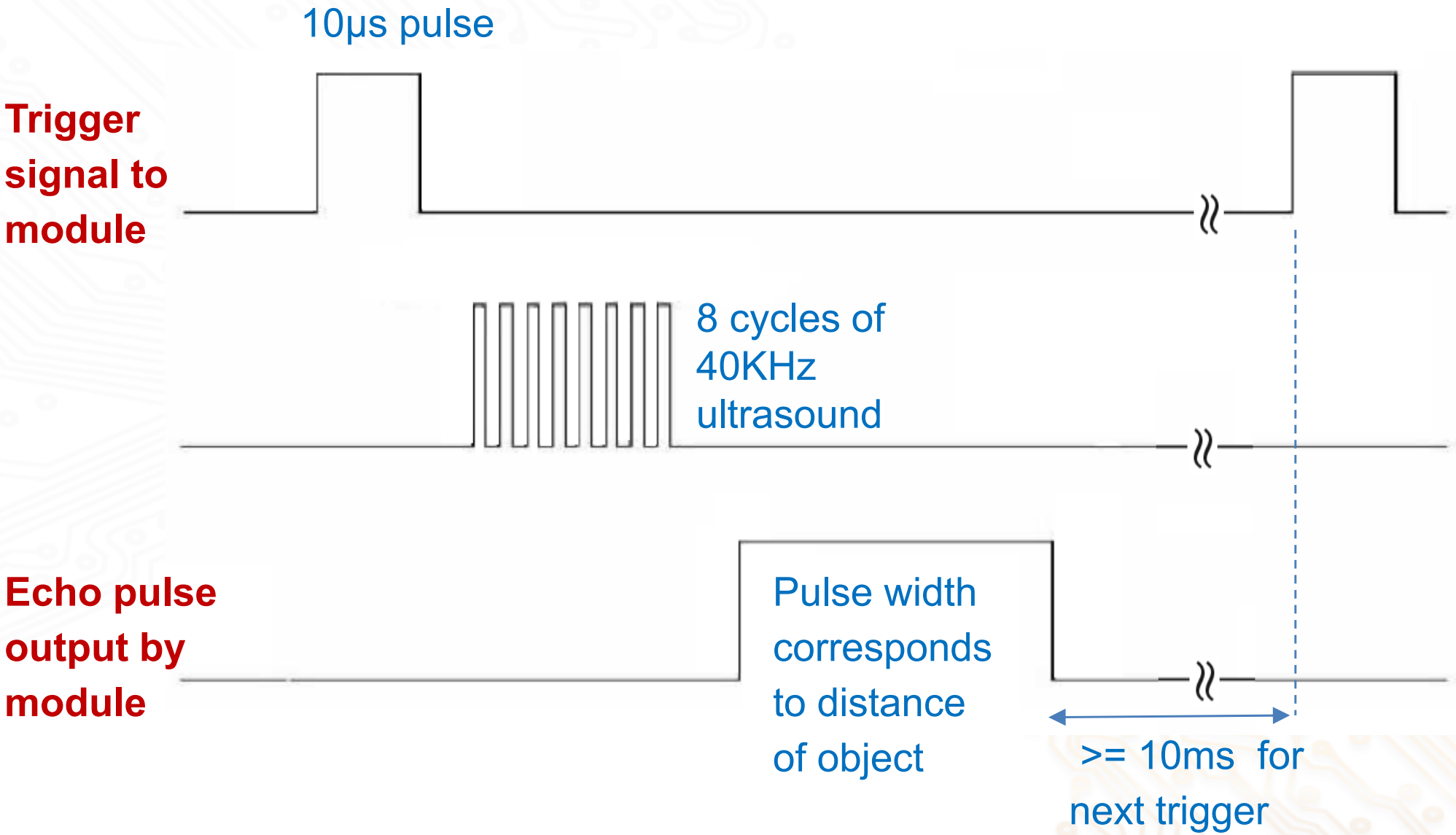
Consists of 4 interface pins

- Vcc: +5V
- Gnd: 0V
- Trig: trigger signal to emit output pulses
- Echo: reflected signal detected





Timing Diagram



Range Calculation

Distance can be calculated by using the formula:

Distance = Echo pulse width * Sound velocity / 2

Sound velocity = 343 meters/sec @ $20^{O}C$

Echo output will be terminated(reset) after 36ms if there is no reflected signal received

- i.e., no object detected, or object farther than 4 meters

Phased Array Ultrasonics

An ultrasonic module can only send the beam in a fixed direction

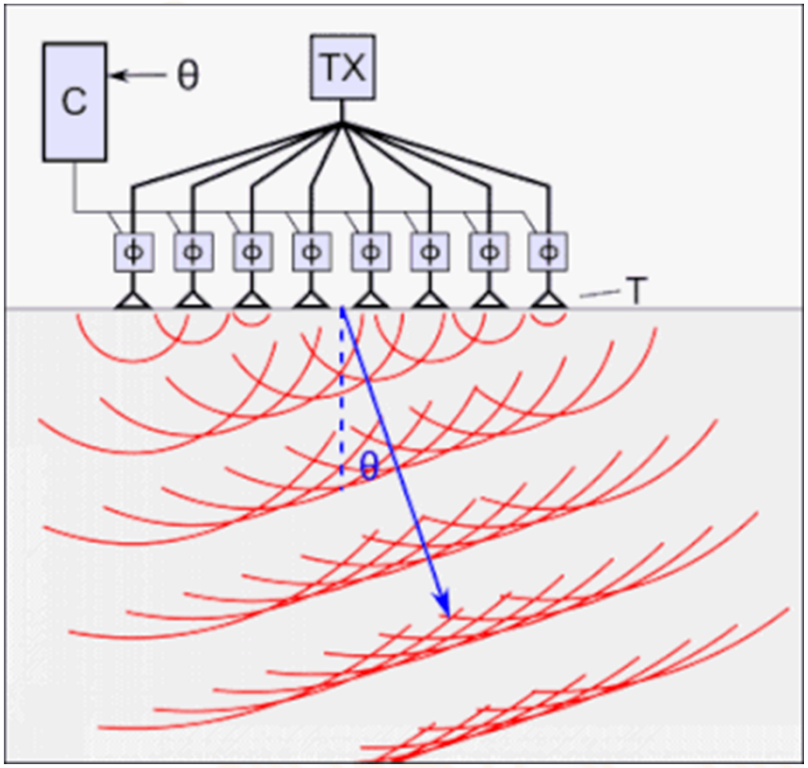
- can be mounted on a movable arm to sweep the beam through the area of interest

A more advanced method

- use multiple Ultrasonic modules arranged in an array
- delay is added between beams emitted by each module

Resultant beam can be directed at different angle

- and swept electronically without moving the array



(Source: Wikipedia)

Summary

Ultrasonic devices are commonly used in applications that needs to scan and measure distances between object.

Emitting beam of signal running at frequency above 20KHz.

Operated by measuring the time taken for the reflected beam to be detected.

Phased array arrangement enables the beam to be focused in different direction without physically moving the devices.